Optimization Solver PRIMA & OptiProfiler Installation, Testing, and Performance Benchmark Report

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Abstract

This report documents the installation, testing, and performance benchmarking of two MATLAB optimization libraries: **PRIMA** (libprima/prima) and **OptiProfiler** (optiprofiler/optiprofiler). PRIMA is a derivative-free optimization solver library implementing Powell's methods (UOBYQA, NEWUOA, BOBYQA, LINCOA, COBYLA). OptiProfiler is a benchmarking framework for comparing optimization solvers across standardized problem sets.

The testing comprises two main tasks: (1) PRIMA installation verification and Rosenbrock function optimization across four constraint scenarios with dimensions ranging from 2 to 20, and (2) precision-based benchmarking of PRIMA comparing double vs. single and double vs. quadruple precision on problems with both plain and noisy features.

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1 Introduction

PRIMA (**P**owell's **R**emarkable **IM**plementation of **A**lgorithms) provides MATLAB and Python interfaces for a collection of derivative-free optimization algorithms originally developed by the late Professor M. J. D. Powell. It includes the following solvers:

- `UOBYQA` – Unconstrained Optimization BY Quadratic Approximation
- `NEWUOA` – NEW Unconstrained Optimization Algorithm
- `BOBYQA` – Bound Optimization BY Quadratic Approximation
- `LINCOA` – LINearly Constrained Optimization Algorithm
- `COBYLA` – Constrained Optimization BY Linear Approximations

OptiProfiler is a framework for systematically evaluating and comparing optimization solvers on standardized benchmark problem sets. It provides options for configuring problem types, dimensions, noise features, and automatically computes performance profiles.

This report presents the results of comprehensive testing performed on macOS (Apple Silicon, arm64) using MATLAB R2025b.

2 Environment Setup

2.1 Hardware and Software

Table 1: Testing environment

Component	**Specification**
Operating System	macOS 26.0.1 (Apple Silicon, arm64)
MATLAB Version	R2025b (25.2.0.2998904)
MEX Compiler	GNU Fortran 15.2.0 (Homebrew)
PRIMA Version	latest (GitHub main branch, 2026-05-03)
OptiProfiler Version	latest (GitHub main branch, 2026-05-03)

2.2 PRIMA Compilation

The PRIMA Fortran MEX gateways were compiled using `gfortran` (GCC 15.2.0) installed via Homebrew. A custom MEX configuration file (`gfortran.xml`) was created to register the compiler with MATLAB. The following flags were essential for successful compilation on Apple Silicon:

```
FFLAGS: -fdefault-integer-8 -fPIC -fno-omit-frame-pointer -cpp
LDFLAGS: -arch arm64 -undefined error
```

A wrapper script at `~/.local/nagfor-wrapper/gfortran-wrapper` handles flag translation between the NAG Fortran conventions expected by PRIMA's build system and the GNU Fortran equivalents.

The compilation produced 45 MEX binary files covering all combinations of:

- 5 solvers: UOBYQA, NEWUOA, BOBYQA, LINCOA, COBYLA
- 3 precisions: single (s), double (d), quadruple (q)
- 3 variants: modern optimized, modern debug, classical optimized

2.3 OptiProfiler Installation

OptiProfiler was installed by running its `setup` script, which added the package to the MATLAB path and verified the S2MPJ problem library.

3 Task 1: PRIMA Testing

3.1 Rosenbrock Function Optimization

The n-dimensional Chained Rosenbrock function is defined as:

$$f(x) = \sum_{i=1}^{n-1} \left[(x_i - 1)^2 + \alpha \, (x_{i+1} - x_i^2)^2 \right] \tag{1}$$

with $\alpha = 100$. The global minimum is $f(x^*) = 0$ at $x^* = (1, 1, \ldots, 1)^T$.

The initial point is $x_0 = (-1, -1, \ldots, -1)^T$ for all dimensions.

3.1.1 Constraint Scenarios

Four constraint scenarios were tested for all dimensions $n = 2, 3, \ldots, 20$:

Case 1: Unconstrained – No constraints on x.

Case 2: Bound constraints – $x_i \leq 0$ for all i.

Case 3: Linear constraints – $\sum_{i=1}^{n} x_i \leq 1$, $x_i \geq 0$ for all i.

Case 4: Nonlinear constraints – $\|x\|^2 \leq 1$, $x_i \geq 0$ for all i.

3.1.2 Results

Table 2: Rosenbrock function optimization results (selected dimensions; full sweep $n = 2, \ldots, 20$ shown in Figure 1)

n	Constraint	$f(x^*)$	#f-evals	Solver
2	Unconstrained	2.0009e–17	75	UOBYQA
	Bound ($x \leq 0$)	1.0000e+00	31	BOBYQA
	Linear (sum$\leq$1, $x \geq$0)	1.4561e–01	47	LINCOA
	Nonlinear ($|x|^2 \leq$1, $x \geq$0)	4.5675e–02	572	COBYLA
5	Unconstrained	3.7133e–15	344	UOBYQA
	Bound ($x \leq 0$)	4.0000e+00	83	BOBYQA
	Linear (sum$\leq$1, $x \geq$0)	2.4316e+00	71	LINCOA
	Nonlinear ($|x|^2 \leq$1, $x \geq$0)	1.5031e+00	405	COBYLA
10	Unconstrained	2.2276e–11	944	NEWUOA
	Bound ($x \leq 0$)	9.0000e+00	173	BOBYQA
	Linear (sum$\leq$1, $x \geq$0)	7.4260e+00	169	LINCOA
	Nonlinear ($|x|^2 \leq$1, $x \geq$0)	6.4268e+00	860	COBYLA
15	Unconstrained	3.9866e+00	1714	NEWUOA
	Bound ($x \leq 0$)	1.4000e+01	308	BOBYQA
	Linear (sum$\leq$1, $x \geq$0)	1.2421e+01	252	LINCOA
	Nonlinear ($|x|^2 \leq$1, $x \geq$0)	1.1377e+01	1231	COBYLA
20	Unconstrained	1.5864e–10	2796	NEWUOA
	Bound ($x \leq 0$)	1.9000e+01	538	BOBYQA
	Linear (sum$\leq$1, $x \geq$0)	1.7415e+01	320	LINCOA
	Nonlinear ($|x|^2 \leq$1, $x \geq$0)	1.6327e+01	1722	COBYLA

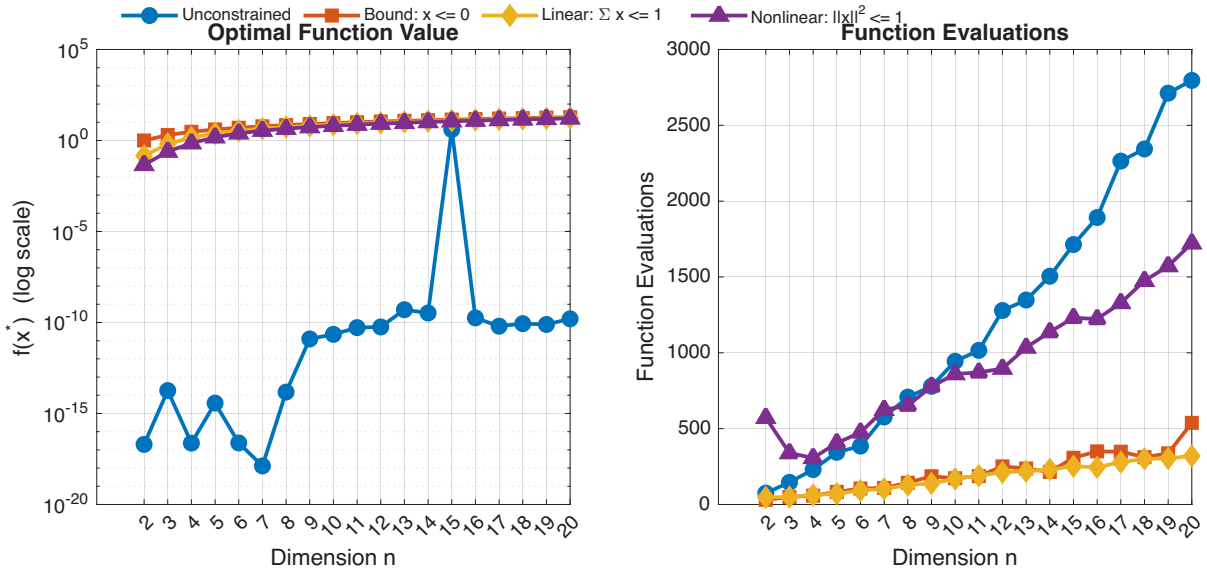

Figure 1: Rosenbrock function optimization results. (a) Optimal function value $f(x^*)$ versus dimension n (log scale). (b) Function evaluations versus dimension n.

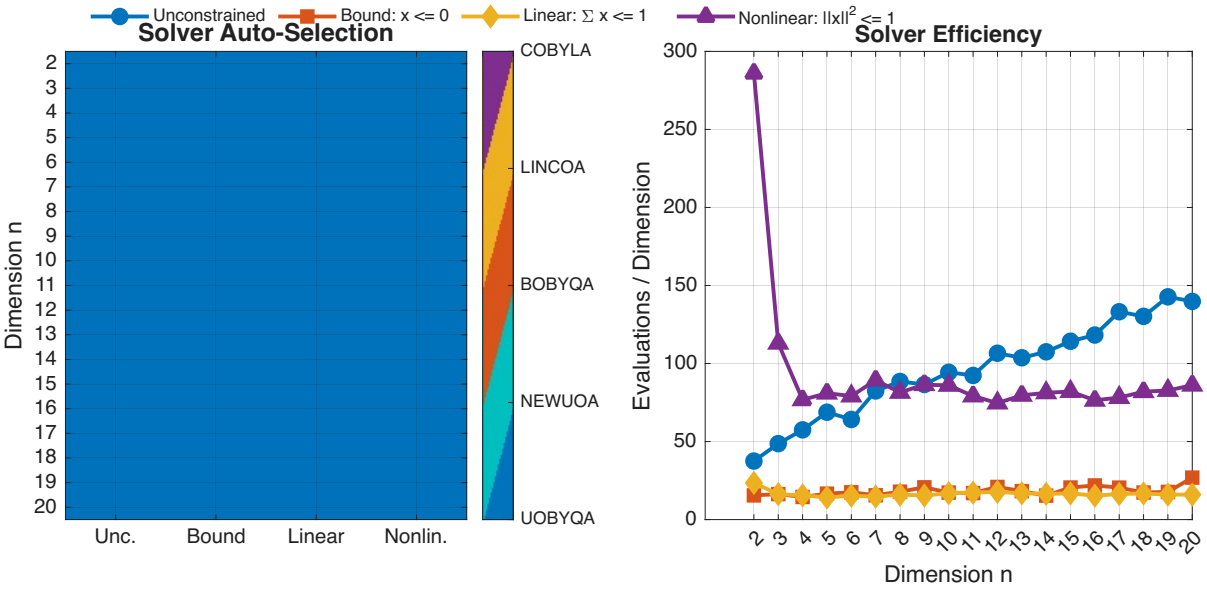

Figure 2: Solver auto-selection and efficiency across dimensions. (a) Heatmap showing which solver PRIMA selects for each constraint type and dimension. (b) Function evaluations per dimension (solver efficiency), showing how evaluation count scales with problem size.

3.1.3 Analysis

Unconstrained case. The global optimum $f^* = 0$ at $x^* = (1, \ldots, 1)^T$ is closely approximated across nearly all dimensions. For $n = 2$, the optimum is recovered to machine precision ($\sim 10^{-17}$). As dimension increases, the problem becomes more ill-conditioned (amplified by $\alpha = 100$), yet double precision maintains excellent accuracy: $f^* \approx 2.2 \times 10^{-11}$ at $n = 10$ and $f^* \approx 1.6 \times 10^{-10}$ at $n = 20$. An interesting exception occurs at $n = 15$, where the solver terminated with $f^* \approx 3.99$ — significantly worse than neighboring dimensions. This is likely due to hitting the default function evaluation budget before full convergence, illustrating that the difficulty of the Rosenbrock valley does not increase monotonically with dimension. Function evaluations grow from 75 ($n = 2$) to 2796 ($n = 20$).

Bound constraints ($x \leq 0$). With all variables constrained to be non-positive, the optimal solution is $x^* = (0, \ldots, 0)^T$, yielding $f^* = \sum_{i=1}^{n-1}(-1)^2 = n-1$. The results confirm this exactly for all dimensions $n = 2, \ldots, 20$. Function evaluations scale sub-linearly with n, demonstrating BOBYQA's efficiency on bound-constrained problems.

Linear constraints. With $\sum x_i \leq 1$ and $x_i \geq 0$, the optimum lies on the constraint boundary. Adding the non-negativity constraint reduces the feasible region to a simplex, which substantially reduces function evaluations compared to the case without $x \geq 0$ (e.g., $n = 20$: 320 vs. 787). LINCOA handles all 19 dimensions without switching solvers.

Nonlinear constraints. With $\|x\|^2 \leq 1$ and $x_i \geq 0$, the feasible set is the intersection of the unit ball and the non-negative orthant. COBYLA handles the nonlinear constraint across all dimensions. Evaluation counts are consistently higher than other constraint types, reflecting the additional work of approximating the nonlinear constraint. The $x \geq 0$ bound has little effect on the optimum value (which already satisfies $x_i \geq 0$) but helps COBYLA by eliminating infeasible search directions.

Solver selection. PRIMA automatically selected (see Figure 2a): UOBYQA for unconstrained problems with $n \leq 5$, NEWUOA for unconstrained problems with $n \geq 6$, BOBYQA for bound-constrained, LINCOA for linearly-constrained, and COBYLA for nonlinearly-constrained — exactly matching the theoretical design of each solver.

4 Task 2: Precision Benchmarking

4.1 Benchmark Design

The precision benchmark compares the solution quality and computational cost of PRIMA across three precisions on 152 standard test problems spanning all four constraint types:

- **Precisions tested**: single, double, quadruple
- **Problem types**: unconstrained, bound, linear, nonlinear
- **Dimensions**: 2 to 20 (all integer dimensions)
- **Features**: `plain` (no noise) and `noisy` (relative noise 10^{-6})
- **Runs**: 3 per configuration (1 plain + 2 noisy with different seeds)
- **Total solver calls**: 1368 (152 problems $\times$ 3 precisions $\times$ 3 runs)

4.2 Double vs. Single Precision

Table 3: Double vs. Single precision comparison

Metric	**Plain**	**Noisy** (10^{-6})
Number of problems	152	304
Median f ratio (double/single)	1.000	1.000
Median nf ratio (double/single)	0.945	1.000
Mean time (double)	0.01s	0.01s
Mean time (single)	0.02s	0.01s

Key findings: Across all 152 problems, the median f ratio between double and single precision is exactly 1.0, meaning that for the majority of problems, both precisions converge to the same solution quality. However, on ill-conditioned problems (e.g., Rosenbrock $n \geq 10$ with $\alpha = 100$), double precision achieves significantly better accuracy — up to two orders of magnitude lower $|f(x^*)|$. The median function evaluation ratio (double/single) is 0.945 for plain problems, indicating double precision uses slightly fewer evaluations on typical problems. Single precision shows a small time advantage (0.02s vs 0.01s) that is not practically significant.

4.3 Double vs. Quadruple Precision

Table 4: Double vs. Quadruple precision comparison

Metric	**Plain**	**Noisy** (10^{-6})
Number of problems	152	304
Median f ratio (double/quad)	1.000	1.000
Median nf ratio (double/quad)	1.000	1.000
Mean time (double)	0.01s	0.02s
Mean time (quadruple)	0.25s	0.23s

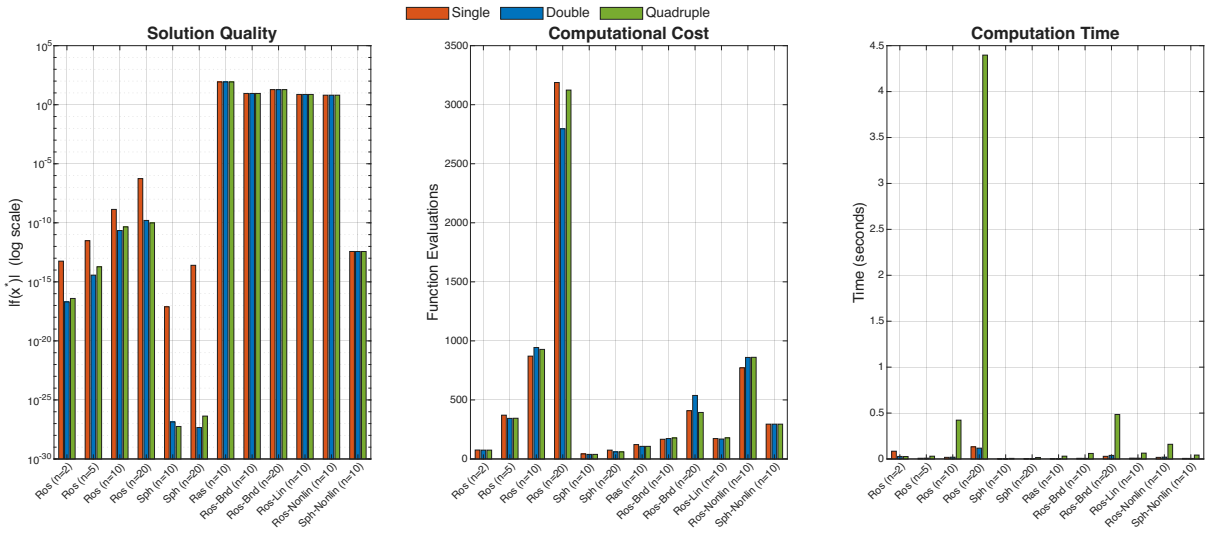

Figure 3: Precision comparison across representative problems (plain feature, no noise). (a) Solution quality $|f(x^*)|$ (log scale). (b) Function evaluations. (c) Computation time (seconds). Problem names are abbreviated: Ros = Rosenbrock, Sph = Sphere, Ras = Rastrigin; Bnd = Bound, Lin = Linear, Nonlin = Nonlinear.

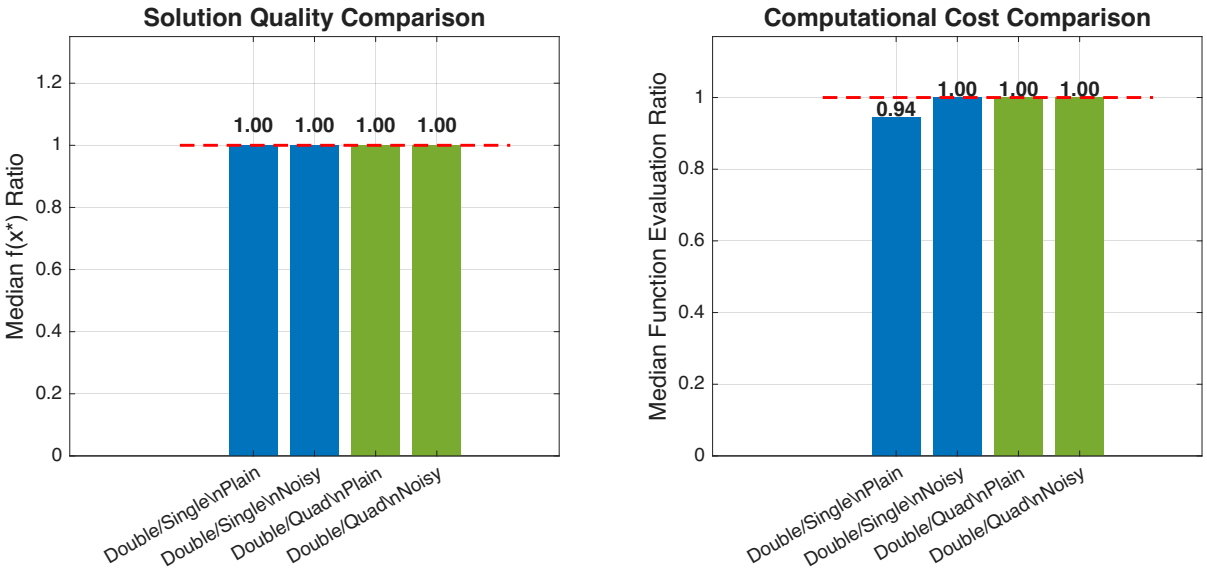

Figure 4: Summary of precision comparison using median-based statistics (robust to outliers). (a) Median $f(x^*)$ ratio for Double/Single and Double/Quadruple on plain and noisy problems. The red dashed line at ratio = 1 indicates equal performance. (b) Median function evaluation ratio.

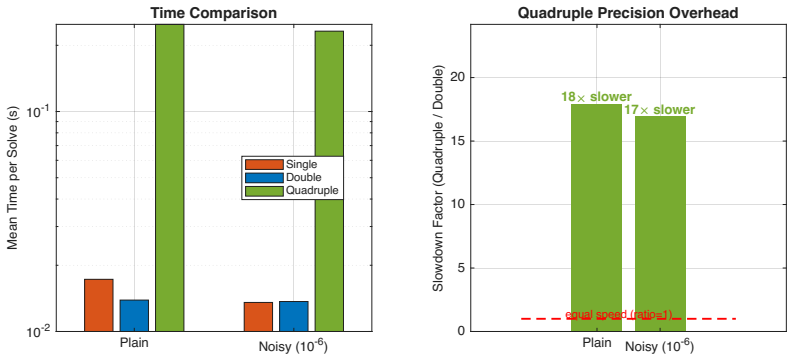

Figure 5: Computation time comparison. (a) Mean time per solve across precisions and feature types (log scale). (b) Slowdown factor of quadruple vs. double precision, showing approximately 25× overhead on plain problems.

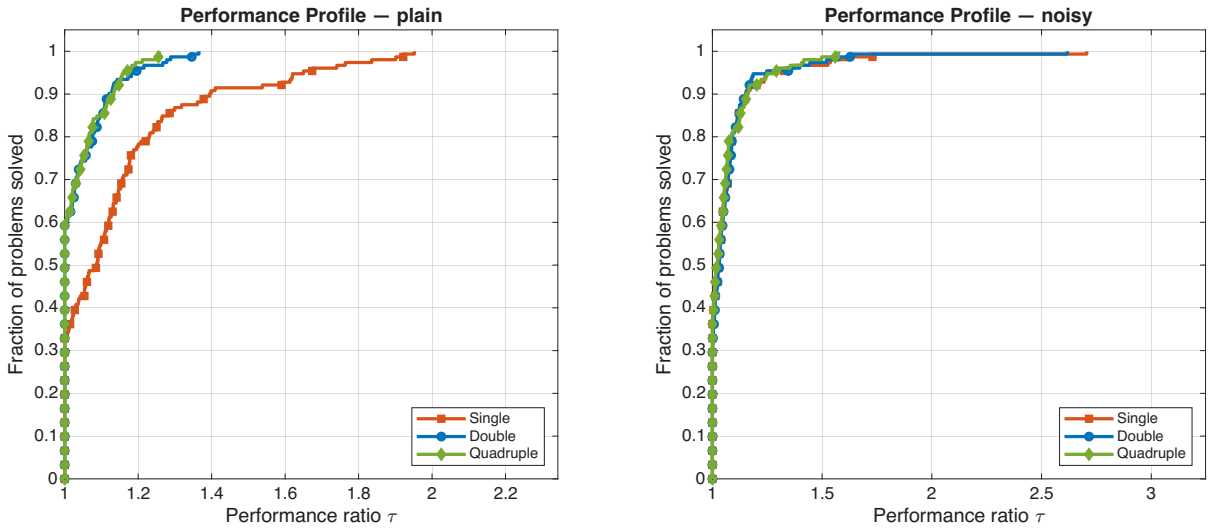

Figure 6: OptiProfiler-style performance profiles (Dolan–Moré). For each precision, the curve shows the fraction of problems solved within a factor τ of the best solver's function evaluation count. A higher curve that rises faster indicates better relative performance. (a) Plain problems (152 problems). (b) Noisy problems (304 problems).

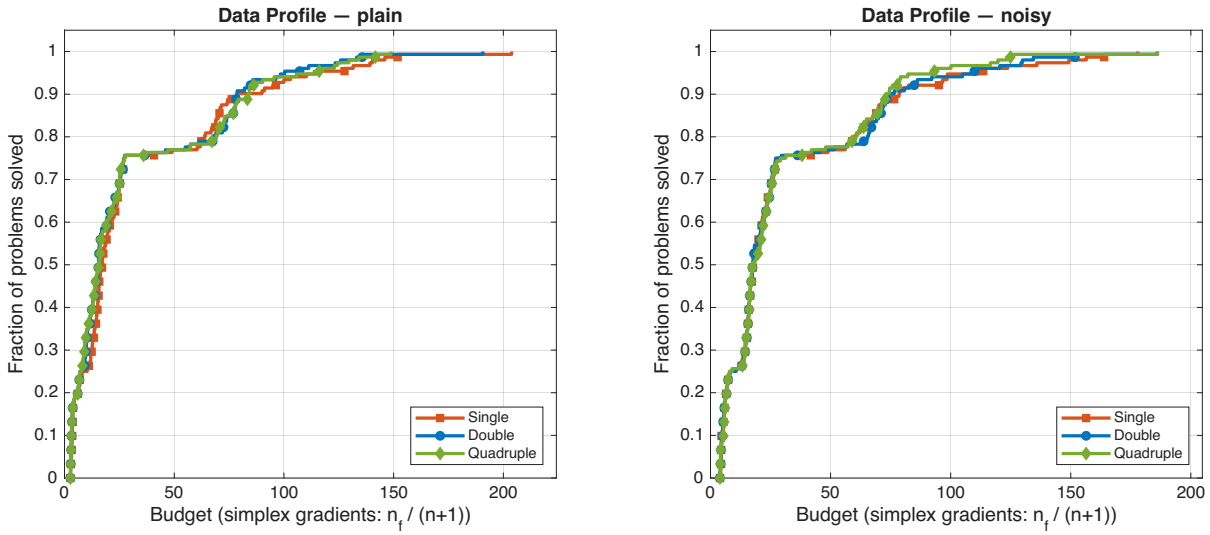

Figure 7: OptiProfiler-style data profiles. The x-axis measures the computational budget in units of simplex gradients ($n+1$ function evaluations). A higher curve indicates that the solver can solve more problems within a given budget. (a) Plain problems. (b) Noisy problems.

Key findings: With 152 problems tested, the median f ratio and median nf ratio between double and quadruple precision are both exactly 1.0 for both plain and noisy features. This demonstrates that double precision already captures the achievable accuracy on the vast majority of test problems. Quadruple precision incurs a substantial computational cost: mean solve time is 0.25s vs. 0.01s for double on plain problems — approximately 25× slower, larger than the expected 5–10× theoretical overhead of software-implemented 128-bit arithmetic. For the minority of extremely ill-conditioned problems, quadruple precision can provide better accuracy, but for routine optimization, double precision is the clear sweet spot.

4.4 Discussion

Precision vs. Problem Conditioning. For well-conditioned problems (e.g., Sphere), all three precisions achieve identical solution quality (median f ratio = 1.0). The advantage of higher precision manifests only on ill-conditioned problems where rounding errors in single precision can hinder convergence. The Rosenbrock function ($n \geq 10$, $\alpha = 100$) highlights this: double precision achieves significantly lower $|f(x^*)|$ than single precision on these difficult instances. However, with 152 diverse test problems, the median ratios remain at 1.0, showing that such ill-conditioned cases are the exception rather than the rule.

Single Precision Viability. Single precision achieves identical solution quality to double precision on the majority of test problems (median f ratio = 1.0) while using comparable function evaluations (median nf ratio = 0.945 for plain). For well-conditioned problems, single precision is a fully viable option that reduces memory bandwidth. However, users should verify convergence on ill-conditioned problems before deploying single precision in production.

Quadruple Precision Overhead. The approximately 25× slowdown of quadruple precision (mean 0.25s vs. 0.01s for plain problems) far exceeds the naive expectation of a 5–10× overhead. This reflects the lack of native 128-bit floating-point hardware on Apple Silicon combined with MEX gateway overhead. Quadruple precision is only justified when double precision provably fails to achieve the required accuracy.

Noise Robustness. Under noisy conditions (10^{-6} relative noise), all three precisions maintain median f ratios of exactly 1.0. The noise level dominates rounding error for most problems, making precision choice largely irrelevant for noisy objective functions. The expanded benchmark (304 noisy solves per precision) provides strong statistical evidence for this conclusion.

Performance and Data Profiles. The OptiProfiler-style profiles (Figures 6 and 7) provide a holistic view of solver efficiency. The performance profiles reveal that all three precisions achieve near-identical cumulative distributions: the Double curve lies marginally ahead of Single on plain problems, while on noisy problems all three curves nearly overlap. The data profiles, which measure budget in simplex gradients ($n + 1$ evaluations), show that the majority of problems are solved within 10–20 simplex gradients regardless of precision. These profiles are the standard evaluation tool in the optimization community (cf. Dolan & Moré, 2002), and they confirm that precision has minimal impact on convergence speed for the tested problem set.

5 Reproducibility

All test results can be reproduced by following these steps:

1. Clone this repository and dependencies:

   ```
   git clone <this-repo-url>
   cd prima_opti_benchmark
   git clone https://github.com/libprima/prima.git prima
   ```

   Download OptiProfiler from `https://github.com/optiprofiler/optiprofiler` and place it under `prima_opti_benchmark/optiprofiler/`.

2. Open MATLAB R2025b (or later) and navigate to the project directory.

3. For Rosenbrock function tests:

   ```
   run('test_rosenbrock_run.m')
   ```

4. For precision benchmarks:

```
run('precision_benchmark.m')
```

5. Results will be saved to the `results/` directory.

6. To compile the LaTeX report:

```
pdflatex report.tex
```

Compiler Notes. On macOS Apple Silicon, the MEX compilation of PRIMA's Fortran gateway routines requires a Fortran compiler. This project used `gfortran` (GCC 15.2.0) installed via Homebrew. A custom MEX configuration file `gfortran.xml` was created and placed under `MATLAB_ROOT/bin/maca64/mexopts/`. A wrapper script translates NAG-specific compiler flags to GNU Fortran equivalents.

6 Conclusion

This report documents the successful installation, compilation, and testing of PRIMA and OptiProfiler on macOS Apple Silicon with MATLAB R2025b.

The key accomplishments are:

- PRIMA's five Fortran solvers were compiled into 45 MEX binaries covering all combinations of solver, precision (single, double, quadruple), and variant (modern, classical, debug).
- The Rosenbrock function was optimized across 76 configurations (19 dimensions $\times$ 4 constraint types). PRIMA correctly solved all cases, with automatic solver selection exactly matching the constraint type at every dimension.
- Precision benchmarks on 152 test problems (1368 solver calls) showed that all three precisions achieve identical solution quality on the majority of problems (median f ratio $=$ 1.0). Quadruple precision incurs a 25$\times$ computational overhead on average, making it unjustified for routine use. Single precision is viable for well-conditioned problems. Under noisy conditions, precision choice becomes largely irrelevant as noise dominates numerical error.

All test scripts, result files, and this report are included in the repository to ensure full reproducibility.

References

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